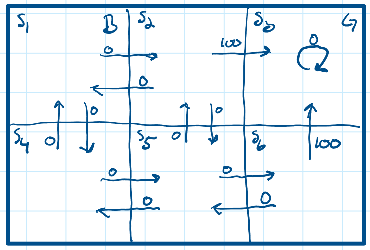
CS_5080_Lecture_4

Thursday, January 29, 2026 4:37 PM

Starting where lecture 3 left off



Dynamics of an RL Problem

① State-action-state
or transition table

From state

	actions				
	↑	↓	←	→	⟲
s_1	NA	s_4	NA	s_2	NA
s_2					
s_3					
s_4					
s_5					
s_6					

② State-action-immediate reward table
or Reward table

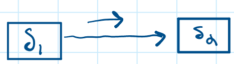
From state

	action				
	↑	↓	←	→	⟲
s_1	NA	0	NA	0	NA
s_2	NA	0	0	100	NA
s_3	NA	NA	NA	NA	0
s_4	0	NA	NA	0	NA
s_5	0	NA	0	0	NA
s_6	100	NA	0	NA	NA

Non-determinism (CS4700)

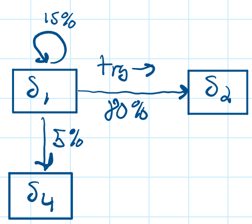
Actions

→ : go left action



instead use

Try→ : Try to go left action



or could use 100% → s_2 and 0%, 0%

Reward Non-determinism

For the same action at the same state the rewards are stochastic

Most complex nondeterministic case

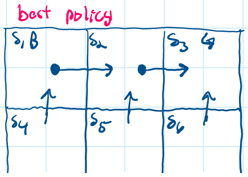
a combination of these two

Goal of RL

To Learn a better policy, possible the ideal/best

There may be many or infinite possiblities but "one" best policy may be

best policy



what is the best action to perform in each state

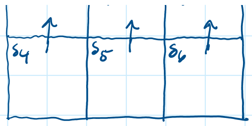
Policy π

initial policy π_0

$$\pi_0 = p(a|s) \quad \forall s \in S$$

assume an equi-possible policy [or it may be random]
all actions possible in a state are equally probable
can be written as a table in the simple case

	action				
state	↑	↓	←	→	⟲
s_1	0.5	0	0	0.5	0
s_2					
s_3					
s_4					
s_5					
s_6					



to perform in each state

Policy maye include offline knowledge

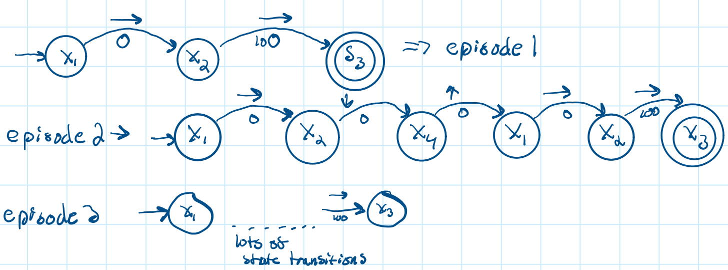
Learned "best" policy

state	action
s_1	→
s_2	→
⋮	
s_6	

Reinforcement learning

given the dynamics of the problem and π_0 (random, equi-prob, expert-based) learn a "best" policy.

given the dynamics of the "maze" use equi-prob policy (say)



need to find out how good state s_t is for the RL agent wants to maximize the cumulative immediate rewards

Return G_t of state s_t

$$G_t = R_{t+1} + R_{t+2} + \dots + R_T$$ [sum of immediate rewards]
simple cumulative reward

$G_{1,1}$ ← episode   for this maze all $G_t = 100$
would need to track steps to make this learn

could use this:

$$\text{Return} \Rightarrow G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots + \gamma^{T-t} R_T$$

γ ← discount factor gamma

where gamma $= 1 \geq \gamma \geq 0$

assume $\gamma = 0.9$

$$G_1 = 0 + \gamma * 100 = 0 + 90 = 90$$

$$G_2 = 0 + \gamma * 0 + \gamma^2 * 0 + \gamma^3 * 0 + \gamma^4 * 100 = 65.61$$

$$G_3 = 0.002656$$

Value of a state

$V_{episode_1}(x_1) = G_1$ episode 1

In general $V(S_t) = G_t$ in some episode

What is the generic value of a state in an episodic RL problem?

$$V_\pi(x) = \mathbb{E}\left[G_t \mid S_t = x\right] \quad x \in \mathbb{S} \quad \Rightarrow \quad \frac{1}{\#\text{episodes}}\left[R_{t+1} + \gamma R_{t+2} + \dots\dots\right]$$

$\uparrow$ Policy (under π)

$\uparrow$ mean value (under $\mathbb{E}$)

Naive algor to learn $V_\pi(s)$

For every state s $\begin{bmatrix} \text{no\_visits}[s] = 0 \\ V(s) = 0 \end{bmatrix}$

for i = 1 to no_episodes

start episode$_i$ from a start state

finish episode$_i$

for $t = 1$ to T_i

$$G_{t,i} = R_{t+1,i} + R_{t+2,i} + \dots\dots$$

$$V(S_t) = \frac{\text{no\_visits} * V(S_t) + G_{t,i}}{\text{no\_visits} + 1}$$

Issues with naive algor

- in big environments, this is very very slow
- convergence may not happen
- deterministic
- may not visit all states